

Q2: End-to-End Vendor Procurement System

A complete procurement system for QOSMIC SPACE, covering the full
lifecycle
from internal request to vendor quality tracking

QOSMIC SPACE — Founder's Office Interview
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Live Prototype & Automations

This system is fully operational. The Airtable base, Make.com automations, and Slack alert channel are all live and connected:

- **Airtable base:** [QOSMIC Procurement Base](#) — 5 tables, linked records, Kanban views, Form views
- **Make.com Scenario 1:** [PO Past Promised Delivery Date Alert](#) — polls Airtable every 15 minutes, sends Slack alert when a PO is past due
- **Make.com Scenario 2:** [New High-Value Purchase Request \(₹50,000+\)](#) — sends Slack alert for CEO approval
- **Slack channel:** [#procurement-alerts](#) — receiving live automated alerts

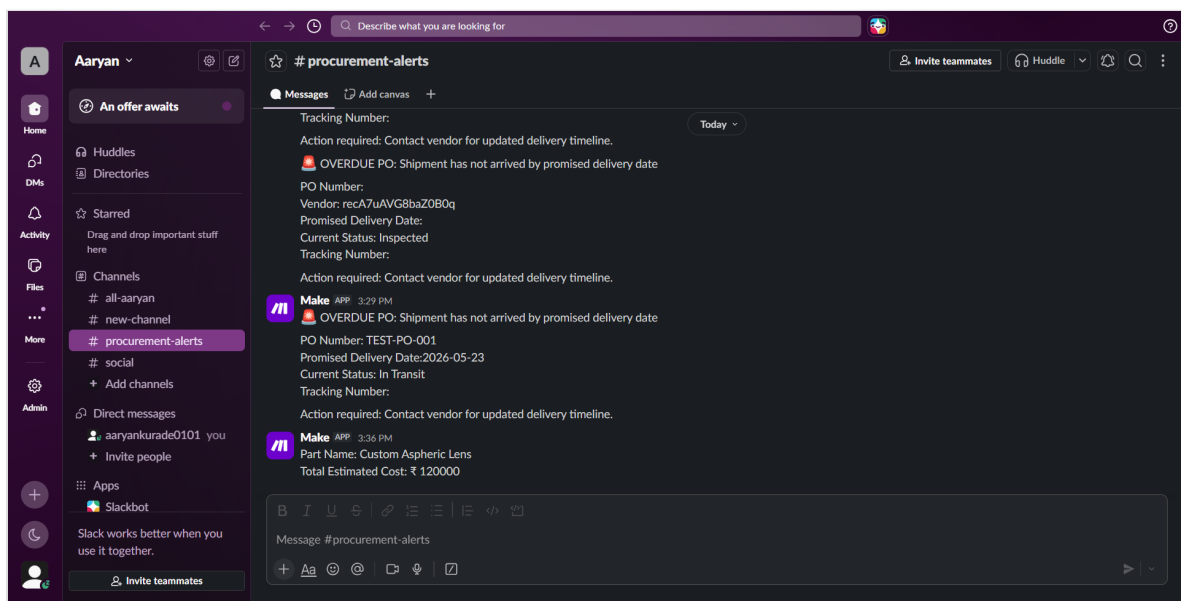


Figure: Slack #procurement-alerts channel showing the PO overdue alert and high-value request alert firing from live Make.com automations.

1. Internal Purchase Request Form

A standardised intake form that captures everything the procurement person needs to source a part. Every field answers a specific downstream question.

Requester Information

Field	Required	Purpose
Requester name	Yes	Who to contact for clarifications
Department/Team	Yes	Optics, Mechanical, Electronics, Systems
Date requested	Yes	Start of procurement clock
Required-by date	Yes	Drives vendor selection (fast vs. standard lead time)
Budget code	Yes	Cost allocation and spend tracking

Part Information

Field	Required	Purpose
Part name	Yes	Human-readable identifier
Part number	Conditional	For reorders: reference previous PO
New part or reorder	Yes	Reorders skip RFQ — go straight to PO
Quantity	Yes	Total units needed
Estimated unit cost	Yes	Enables budget check before approval
Total estimated cost	Yes	Quantity × estimated unit cost

Abridged Form for Reorders

When the part has been procured within the last 6 months, only 5 fields are required: requester, department, date, quantity, required-by date, budget code, and part number (which auto-populates the rest). This makes reordering a ~60-second task.

2. Custom-Machined Parts Pre-Submission Checklist

Catches common specification errors before the request reaches a vendor. Items in Drawing and Material sections are hard gates — submission is blocked if any fail.

Drawing Requirements (Hard Gate)

- Drawing file attached and viewable (PDF or DXF/STEP)
- Fully dimensioned — every feature has at least one dimension callout
- Tolerance callouts on critical dimensions ($\pm 0.1\text{mm}$, $\pm 0.05\text{mm}$, or GD&T)
- Complete thread specs (M6 $\times$ 1.0 - 6H, 12mm depth — not just "M6")
- Revision letter and title block
- Units explicitly stated (mm or inches)

Material Requirements (Hard Gate)

- Material spec includes alloy/grade (6061-T6, not just "aluminium")
- Temper/heat treatment specified if applicable
- Surface treatment specified (clear anodise, hard anodise, none)
- Material spec consistent between drawing and request form

Why These Specific Checks?

Incomplete thread specs (#1 cause of RFQ clarifications): An engineer writes "M6 thread" without specifying pitch or depth. A machinist needs M6 $\times$ 1.0 - 6H, 12mm depth. Without it, they guess.

Square internal corners (#1 manufacturability error): A 90-degree corner cannot be machined with a round endmill. The engineer must specify a corner radius (R1, R2, R3). This is the single most common reason CNC parts come back wrong.

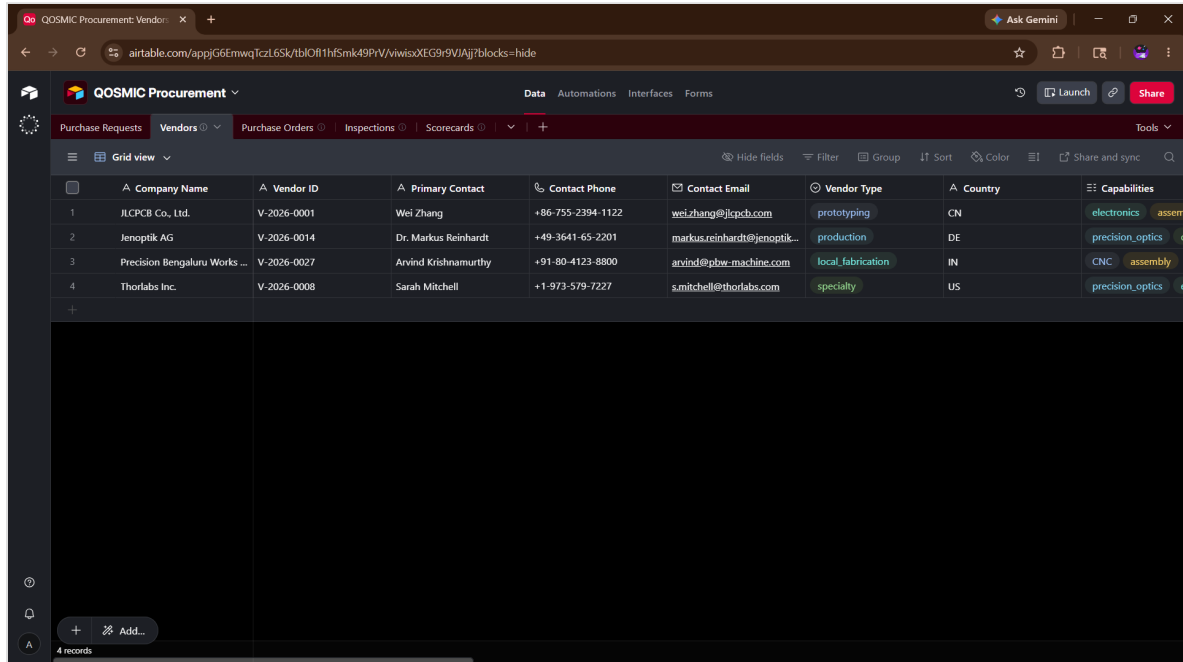
3. Approval Thresholds & Justification

QOSMIC is a seed-stage startup at approximately ₹18L/month burn (~\$18,650/month at ₹96.5/USD). Cash is the constraint. Oversight scales with financial impact.

Purchase Total	Approval Required	Rationale
< ₹5,000 (~\$52)	Self-approved	0.28% of monthly burn. Wrong decision costs less than review overhead.
₹5,000 - ₹50,000 (\$52-\$518)	CTO approval	2.8% of monthly burn. CTO assesses spec and price.
> ₹50,000 (~\$518)	CEO approval	Major optics, long-lead items. Strategic cash commitment.

Emergency exception: Critical-path items where delay costs more than the part itself. CTO can approve up to ₹50K for expedited purchases with documented justification.

4. Vendor Database Schema & Sample Data



The screenshot displays an Airtable interface for a 'Vendors' table. The table is viewed in 'Grid view' and contains 4 records. The columns are: Company Name, Vendor ID, Primary Contact, Contact Phone, Contact Email, Vendor Type, Country, and Capabilities. The records are for JLCPCB Co., Ltd., Jenoptik AG, Precision Bengaluru Works, and Thorlabs Inc.

	Company Name	Vendor ID	Primary Contact	Contact Phone	Contact Email	Vendor Type	Country	Capabilities
1	JLCPCB Co., Ltd.	V-2026-0001	Wei Zhang	+86-755-2394-1122	wei.zhang@jlcpcb.com	prototyping	CN	electronics, assembly
2	Jenoptik AG	V-2026-0014	Dr. Markus Reinhardt	+49-3641-65-2201	markus.reinhardt@jenoptik...	production	DE	precision_optics, CNC
3	Precision Bengaluru Works ...	V-2026-0027	Arvind Krishnamurthy	+91-80-4123-8800	arvind@pbw-machine.com	local_fabrication	IN	CNC, assembly
4	Thorlabs Inc.	V-2026-0008	Sarah Mitchell	+1-973-579-7227	s.mitchell@thorlabs.com	specialty	US	precision_optics

Figure 1: Vendors table — JLCPCB, Jenoptik AG, Precision Bengaluru Works, Thorlabs Inc. with 17 fields each.

5. RFQ Process & Vendor Communication Templates

Vendor Solicitation Guidelines

Part Type	Vendors to Solicit	Rationale
Catalog component	1-2	Published pricing. Second for lead time check.
Custom-machined part	3-5	Competitive. More than 5 = diminishing returns.
Precision custom optics	2-3	Fewer qualified vendors. Compare capability statements.
Electronics / PCBA	2-3	JLCPCB for prototype + local for production.

Three Email Templates

Template A (Initial RFQ): Detailed technical spec for a custom-machined aluminum optical mount. Includes material (6061-T6), surface finish ($Ra \leq 0.8 \mu m$), tolerances ($\pm 0.05 mm$), quantities (5/25/50), shipping terms (FCA), and response deadline (7 business days).

Template B (Follow-up after 5 days): Polite nudge. "Working on the quote and need more time? If you're unable to quote, a brief reply is fine."

Template C (Negotiation at 30%+ over budget): Asks for cost breakdown, offers spec adjustments that save money (relax surface finish on non-functional faces, longer lead time, adjusted payment terms).

6. PO Tracking System

Status Flow

Issued → Acknowledged → In Production → Shipped → In Transit → Delivered
→ Inspected → Accepted / Rejected (with Customs Hold as an intermediate
state for international shipments)

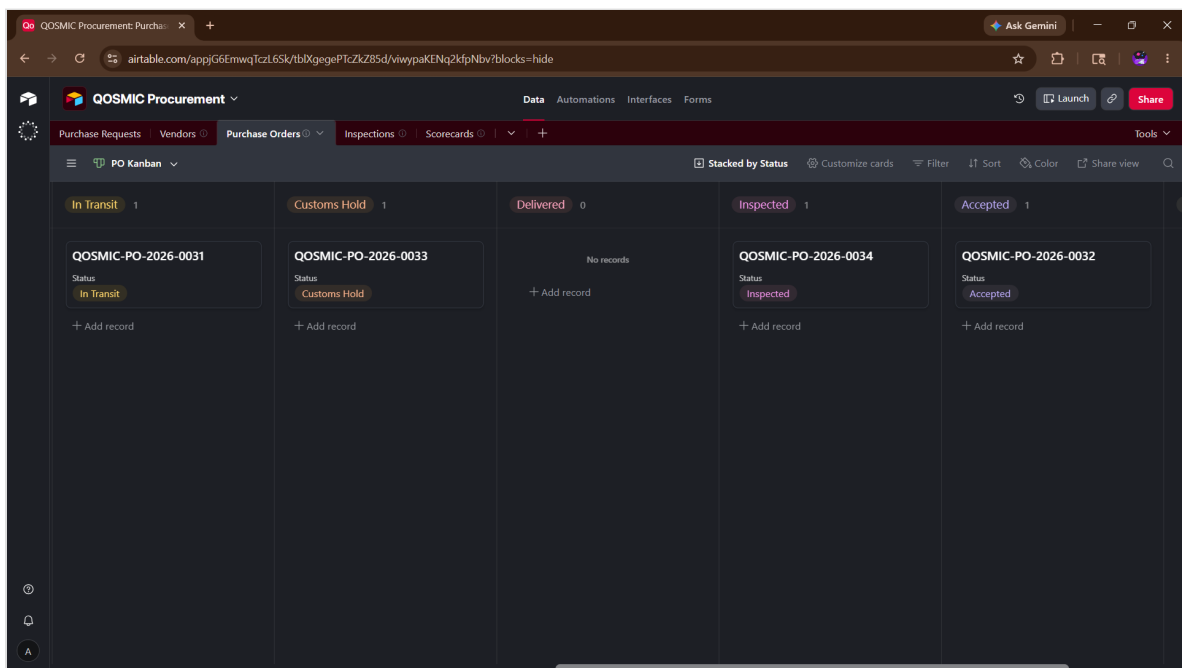
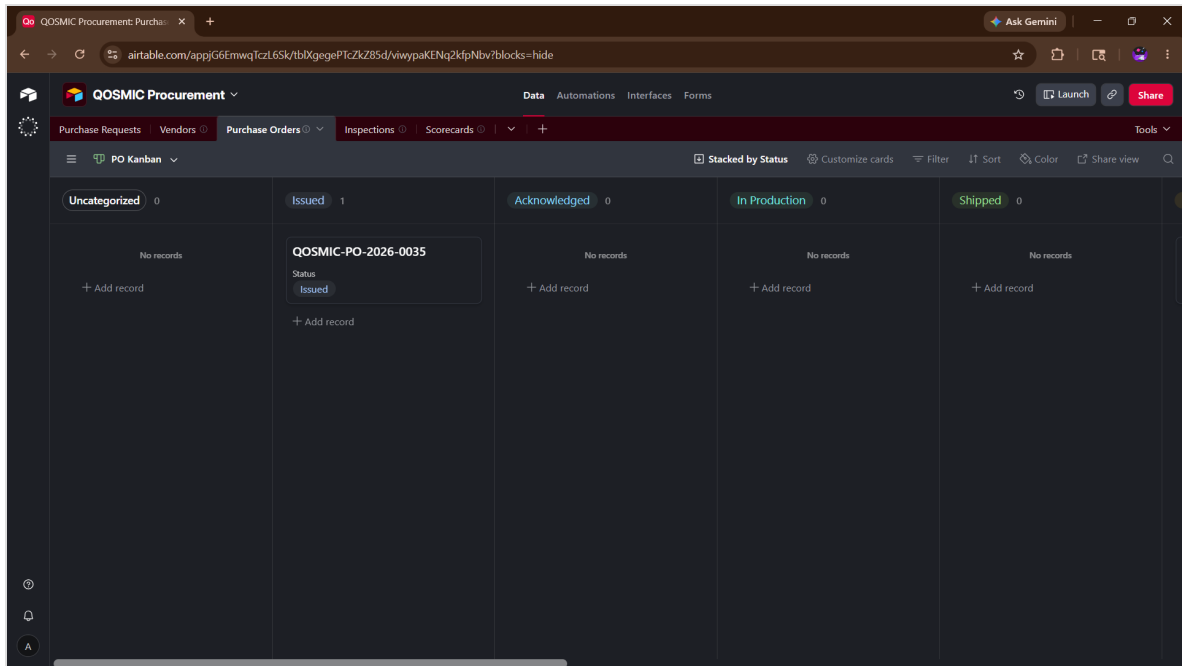


Figure 2a-2b: PO Kanban board showing 5 POs in various stages — Issued (PO-0035), In Transit (PO-0031), Customs Hold (PO-0033), Inspected (PO-0034), Accepted (PO-0032).

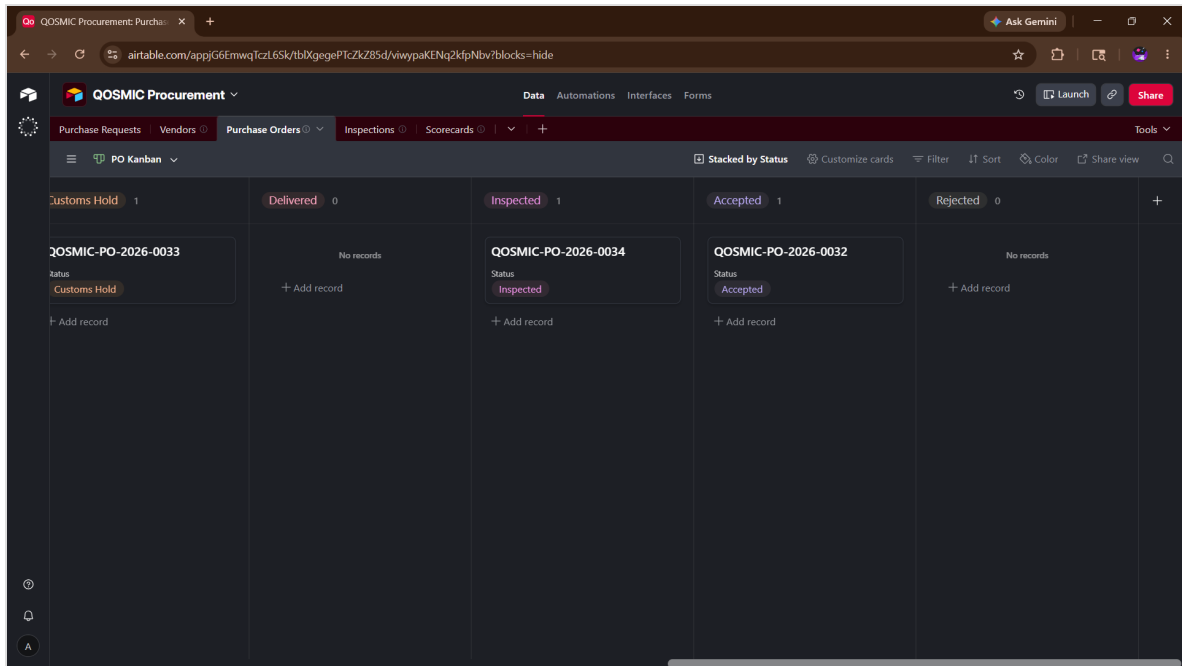


Figure 2c: Customs Hold column showing PO-2026-0033 (custom Jenoptik lens stuck at Bengaluru customs since April 26).

Automated Alert Triggers (2 Live, 5 Manual)

Trigger	Status	Condition
PO not acknowledged	Manual	Status = Issued for >3 business days
Delivery overdue	Automated	Past promised delivery date, status $\neq$ Delivered/Accepted
High-value request	Automated	New request with cost > ₹50,000
Ship date approaching	Manual	5 days before promised ship, no confirmation
No tracking update	Manual	In Transit for >7 days
Customs hold >3 days	Manual	Status = Customs Hold for >3 business days
Quarterly budget exceed	Manual	Vendor spend >120% of quarterly budget

7. Incoming Inspection Checklists

Custom-Machined Parts — Critical (Must Pass)

- Quantity matches PO
- No major defects (cracks, deep scratches, dents)
- Critical dimensions against drawing (calliper, 3 points per dimension)
- Thread gauge check (Go/No-Go)
- Surface finish on optical faces: $Ra \leq 0.8 \mu m$
- Material certification provided

Precision Optical Components

Parameter	Why It Matters	Threshold
Surface quality (scratch-dig)	Scratches scatter 1064nm light, reduce SNR	20-10 for precision, 40-20 for general
Wavefront error	Stacked WFE degrades fiber coupling	$\lambda/10$ at 633nm
Coating reflectivity	0.2% loss per surface $\times$ 15 surfaces = 3%	$\geq 99.5\%$ at 1064nm
Clear aperture	Beam clipping creates diffraction	$\geq 90\%$ of physical diameter

8. Rejection Workflow & Partial Shipments

Workflow

Inspection finds defect → Photograph (3+ angles, scale reference) → Move to quarantine → Send rejection email → 3 business day response window → Resolution (rework / replace / refund) or escalate to CEO.

Rejection Email (Template D)

Specifies measured vs. specified values in a table with FAIL status, attaches photographs and inspection report, and requests one of three resolutions within 3 business days: rework, replacement, or refund.

Partial Shipment Policy

If a batch of 10 parts has 8 good and 2 bad: accept the 8, reject the 2. Vendor replaces the 2. Production doesn't wait for replacements. Rejection rate tracked by unit count, not dollar value (one ₹50K lens failing = 100% unit rejection rate; five ₹200 brackets failing = 100% unit rejection rate — dollar value would hide the bracket problem).

9. Vendor Quality Scorecard

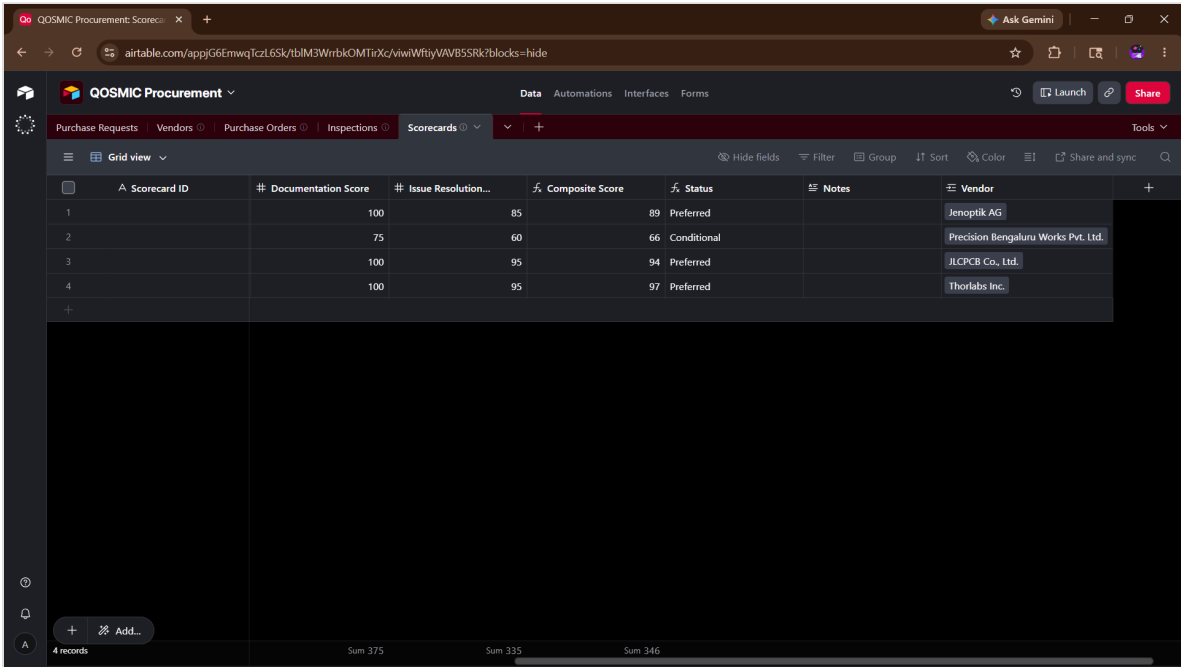


Figure 3: Vendor scorecards — composite scores auto-calculated: Jenoptik 89.0 (Preferred), PBW 67.5 (Conditional), JLCPCB 94.4 (Preferred), Thorlabs 97.1 (Preferred).

7 Metrics & Thresholds

Metric	Target	Weight
On-time delivery rate	≥90%	25%
Quality rejection rate	≤3%	25%
Quote accuracy	≤5%	10%
Lead time consistency	≤7 days	10%
Responsiveness	≤2 business days	10%
Documentation quality	100%	10%
Issue resolution speed	≤5 business days	10%

Composite score: ≥ 85 = Preferred | 70-84 = Approved | 55-69 = Conditional |
<55 = Probation

10. Tool Stack Proposal

Tool	Plan	Cost	Role
Airtable	Free	\$0	Core database, forms, Kanban, linked records
Make.com	Free	\$0	Connects Airtable → Slack, 2 automations live
Slack	Free	\$0	#procurement-alerts channel, live alerts
Gmail	Free	\$0	Vendor communication
Google Drive	Free	\$0	File storage

Total: \$0/month. Why Airtable over alternatives: Notion's automations can't conditionally route based on field values. Excel has no relational integrity. Custom web app violates the "usable by non-developer" constraint.

11. 6-Week Vendor Follow-Up Sequence

Designed so the vendor never feels ambushed. By the time you get firm (week 5), you've already checked in 4 times.

Week	Tone	Key Message
2	Routine	"Just checking in — confirming timeline is still on track. No action needed if all good."
4	Expectation	"Planning our assembly around this delivery. Can you confirm the ship date is still firm?"
5	Firm	"Can you confirm: (1) production completed? (2) confirmed ship date? (3) tracking number?"
6	Escalation	"Shipment was due on [date] and has not arrived. This is impacting our customer delivery. CC'ing [vendor manager]."

12. Airtable Prototype — 5 Sample POs

QOSMIC Procurement: Purcha... x +

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QOSMIC Procurement

Purchase Requests Vendors Purchase Orders Inspections Scorecards

Grid view QOSMIC-PO-2026-0033

PO Number: 1 QOSMIC-PO-2026-0031

Payment Terms: net_45

Tracking Number: DHL-EXP-882011

HS Code: 9002.11

Estimated Duty INR: \$37,200.00

Notes: First article. Held at Bengaluru customs since 2026-04-26. End-user certificate requested by broker. CEO notified.

Vendor: Jenoptik AG

Vendor ID: V-2026-0014

Primary Contact: Dr. Markus Reinhardt

Contact Phone: +49-3641-65-2201

Request: + Add request

Inspections: + Add record

+ Add new field to this table

Start a conversation

Ask questions, keep track of status updates, and collaborate with your team — directly in Airtable.

Invite collaborators

Leave a comment

Sum \$184

QOSMIC Procurement: Purcha... x +

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QOSMIC Procurement

Purchase Requests Vendors Purchase Orders Inspections Scorecards

Grid view

Hide fields Filter Group Sort Color Share and sync

PO Number	Status	Issue Date	Promised Ship Date	Promised Delivery...	Actual Ship Date	Actual Delivery Date	Total Value INR
1 QOSMIC-PO-2026-0031	In Transit	4/1/2026	4/22/2026	4/29/2026	4/24/2026		\$35,
2 QOSMIC-PO-2026-0032	Accepted	3/15/2026	3/18/2026	3/22/2026	3/17/2026	3/22/2026	\$18,
3 QOSMIC-PO-2026-0033	Customs Hold	3/1/2026	4/15/2026	4/25/2026	4/14/2026		\$120,
4 QOSMIC-PO-2026-0034	Inspected	4/10/2026	4/18/2026	4/25/2026	4/18/2026	4/25/2026	\$8,
5 QOSMIC-PO-2026-0035	Issued	5/14/2026	5/21/2026				\$2,

+ Add...

5 records

Sum \$184

Figure 4: PO record expanded view — linked Vendor field (Jenoptik AG), status dropdown, HS code, duty estimate, tracking number, and linked inspection records.

Sample PO Data

PO#	Part	Vendor	Status	Value	Scenario
2026-0031	Custom aluminum mounts	PBW	In Transit	₹35,000	Shipped 2 days late
2026-0032	SM1 lens tubes	Thorlabs	Accepted	₹18,500	Fast-track catalog order, ideal outcome
2026-0033	Custom aspheric lens	Jenoptik	Customs Hold	₹1,20,000	Held at Bengaluru customs since Apr 26
2026-0034	PCB assembly	JLCPCB	Inspected	₹8,200	9 of 10 boards passed, 1 failed
2026-0035	M6 fasteners	PBW	Issued	₹2,800	Self-approved reorder, awaiting acknowledgement

Dashboard metrics: 5 active POs, ₹1.84L total value, 1 overdue (customs hold), 1 rejection this month, 3 vendors engaged.

13. Implementation Cost Estimate

Item	Cost	Notes
Airtable base setup	\$0	Self-built, free plan
Template documents	\$0	Included in this submission
Make.com automations	\$0	Free tier, 2 scenarios live
Scratch-dig paddle	\$165	Edmund Optics #91-291, only if doing in-house optical QC
Digital calliper	\$20-50	0.01mm resolution
Shipping insurance (per shipment)	Varies	0.5-2% of CIF, international only
Customs broker fee (per shipment)	\$31-83	₹2,000-10,000 typical for India

Recurring tooling cost: \$0/month. Total system cost excluding operational shipping/insurance/brokerage: \$0-6.50/month.